[WEEK 4] Count Data and Table Analysis

CS5701 - Quantitative Data Analysis

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Lecture 4 - Housekeeping

- I will start at 11:05 prompt so please be in the Lecture theatre promptly
- Dont forget to check in on SEAtS
- The Lecture recordings will be shared in Brightspace but it can take a few hours to make them available
- We will have time for questions but you can also post them on Brightspace Discussion if you prefer ...



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Week 4 - Learning Outcomes

This week after the lecture, the lab and the independent practice you should be:

- work with count data
- able to use R to run table analysis and interpret the results

This week's material is also covered in Chapter 6 p100 - 107 of Crawley Statistics an Introduction using R.

Example

- Clinicians had a hypothesis that using a shorter but stronger course of antibiotics would be more effective at preventing infection after surgery for mid-facial fractures
- They collected data from 50 patients 28/50 (56%) were prescribed a shortened course
- Of those given a shortened course of anti-biotics 1/28 (3.6%) experienced infective post operative complications.
- 4/22 (18.2%) of those given a longer course of antibiotics experienced infective post operative complications.

They concluded that: "This data supports growing evidence that shorter courses of antimicrobials are appropriate in the surgical management of facial fractures."

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Example - contd

- They submitted this to a journal for publication
- and they got rejected as they did not actually test this hypothesis - just reported the numbers....
- .... and then the statistician got involved

Example - the process of statistical analysis

"To consult the statistician after an experiment is finished is often merely to ask them to conduct a post mortem examination. They can perhaps say what the experiment died of."

Ronald Fisher

The data - antibiotic dose

	short-AB	long-AB
No infection	27	18
Infection	1	4

Is there an improvement??

Example - Antibiotics contd

- An improvement would mean that there is a relationship between the use of ABs and the proportion of patients with infections
- The NULL hypothesis is that there is no relationship between AB use and infection
- The Alternative hypothesis (H1) is that there is a relationship

We have not come across this combination so far.....

Hypothesis Testing - tests introduced so far

So far we introduced hypothesis tests for:

- a mean (e.g. height of children)
- a proportion (e.g. proportion of red sweets in the Gumball machine)
- normality
- equal variances
- equal means

This situation does not fit into any of these...

Count data

- A lot of information comes in the form of **counts**
- This can be the number of students attending a lecture, the number of trees in different neighbourhoods and similar
- For this type of dependent variable it is desirable to model whether the explanatory variable affects the counts for the dependent variable

Lets look at an example...

An example

- We have data on a person's eye colour and their hair colour
- The objective is to test a hypothesis that eye and hair colour are independent

	Blue Eyes	Brown Eyes
Fair Hair	38	11
Dark Hair	14	51

(*) This example is taken from Crawley page 100 onwards.

Example contd

- The table below contains the **observed** frequencies
- The Null model is that hair and eye colour are independent
- Independence is H_0 in this case

	Blue Eyes	Brown Eyes
Fair Hair	38	11
Dark Hair	14	51

Example contd

- The assumption that hair and eye colour are independent is used to compute the **expected frequencies**
- Lets start with calculating the probability that if we randomly select one person from this sample they will have fair hair
- In order to compute these it helps to expand the table to include the row and column totals

	Blue Eyes	Brown Eyes	Row Totals
Fair Hair	38	11	49
Dark Hair	14	51	65
Column Totals	52	62	114

Example contd

What is the probability that a person from this sample will have fair hair?

	Blue Eyes	Brown Eyes	Row Totals
Fair Hair	38	11	49
Dark Hair	14	51	65
Column Totals	52	62	114

Example contd

What is the probability that a person from this sample will have fair hair?

- There are 114 people in the sample, and 49 of them have fair hair
- Therefore the probability of selecting a person with fair hair is $\frac{49}{114}$

	Blue Eyes	Brown Eyes	Row Totals
Fair Hair	38	11	49
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What about the probability that a person from this sample has brown eyes?

Example contd

What is the probability that a person from this sample will has brown eyes?

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This assumes that there is no dependency between these two characteristics

Example contd

- Under the assumption that the two traits (hair and eye colour) are independent then the probability of each of the four combinations is a product of the probabilities for each trait
- For example the probability of sampling one person with blue eyes and fair hair is the product of the probability of blue eyes and probability of fair hair
- From the table $p(\text{eyes} = \text{blue}) = \frac{52}{114}$ and $p(\text{hair} = \text{fair}) = \frac{49}{114}$
- So the **expected** frequency (when we have a sample of 114) is $\frac{52}{114} \times \frac{49}{114} \times 114 = 22.35$
- The **observed** frequency in the table is 38
- ... maybe the traits are not independent!

Calculating the expected frequencies

- When computing the expected counts for each of the cells then sample sizes cancel themselves out
- So to compute the **expected** frequency E for each cell use:

$$E = \frac{R \times C}{G}$$

- where R is the row total, C is the columns total and G is the grand total

Example contd

Lets return to the **observed** frequencies

	Blue Eyes	Brown Eyes	Row Totals
Fair Hair	38	11	49
Dark Hair	14	51	65
Column Totals	52	62	114

$G = 114$, what are the **observed** counts for this table?
Hint: We use

$$E = \frac{R \times C}{G}$$

and there are $R_{\text{fair hair}}, R_{\text{dark hair}}, C_{\text{blue eyes}}, C_{\text{brown eyes}}$

Example contd- expected frequencies

The expected frequencies (assuming independence) are:

	Blue Eyes	Brown Eyes	Row Totals
Fair Hair	22.35	26.65	49
Dark Hair	29.65	35.35	65
Column Totals	52	62	114

Notice: the row, column and grand total are unchanged

Testing for independence

- Given the **expected** and **observed** frequencies we need to test whether there is a significant difference
- This is achieved using a χ^2 or **chi squared** test
- The test statistic for χ^2 (Pearson's) is

$$\chi^2 = \Sigma \frac{(O-E)^2}{E}$$

- This is done in R

Lets see this in the example

χ^2 Contingency table testing in R

- The function `chisq.test` computes it for us
- The input to the function is the table below
- if you compute it manually then the results will differ from R as R applies a "continuity correction".
- this correction can be disabled using `correct=F`

	Blue eyes	Brown eyes
Fair hair	38	11
Dark hair	14	51

The `chisq.test` will return the χ^2 value and the p-value.

Table analysis using R

```
## Crawley example

```{r}
#generating the data
count<-matrix(c(38,14,11,51),nrow = 2)
#printing the data
count
#chi sq test
chisq.test(count)
```
```

```
     [,1] [,2]
[1,]   38   11
[2,]   14   51

        Pearson's Chi-squared test with Yates' continuity correction

data:  count
X-squared = 33.112, df = 1, p-value = 8.7e-09
```

The p-value is very small <0.00001 so we have evidence to reject the NULL hypothesis that there is no relationship between eye colour and hair colour.

Table analysis using R

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data:  count
X-squared = 33.112, df = 1, p-value = 8.7e-09
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```
The p-value is very small <0.00001 so we have evidence to reject the NULL hypothesis that there is no relationship between
eye colour and hair colour.
```

The p-value is interpreted in the same way as with any other hypothesis test. If it is small (p-value<0.05) then what we are seeing in our sample (or something more extreme) is very unlikely to occur if the $H0$ is true...

Fisher's Exact Test

- When contingency tables are based on **small samples** in which **one or more of the expected frequencies** is < 5 χ^2 **should not be used**
- If you try in R to use `chisq.test` in such situations, R **will not stop you** - it still returns a p-value, but with a **warning** that "the approximation may be incorrect". It's easy to miss that warning if you're not looking for it.
- **Fisher's exact** test is to be used instead. This is based on computing the probability of the observed frequencies or more extreme values in the table. `fisher.test`
- The `gmodels` library has a handy function `CrossTable` that is able to use the appropriate test in each case.

Fisher's Exact Test - OPTIONAL

	Col1	Col2	Row Totals
Row1	a	b	a+b
Row2	c	d	c+d
Column Totals	a+c	b+d	n

- Given the table such as the one above
- $$p = \frac{(a+b)!(c+d)!(a+c)!(b+d)!}{a!b!c!d!n!}$$

  needs to be computed for the observed values in the table AND more extreme outcomes
- The function `fisher.test(x)` in R can be used for this test

Example - employment data

We have a data set containing `profession` and `emp.status`

	profession	emp.status
1	a	0
2	b	0
3	b	0
4	b	0
5	a	0
6	b	1
7	b	0
8	a	1
9	a	0
10	b	0
11	a	0
12	a	1
13	c	1
14	c	1
15	c	1
16	c	0
17	c	0

Is there a link between profession and employment status? and how would we test or model this?

Example - Employment data

- If we want to test for independence between profession (a or b or c) and employment status (1 or 0) we can use χ^2
- lets look at the contingency table for this data:

	0	1
a	4	2
b	5	1
c	2	3

Should we use χ^2?

Example - employment data

- We should apply **Fisher's exact** in this case as the expected cell counts are small
- Using `Fisher.test` on this data we obtain a p-value $=0.38$, which confirms H_0 of independence

Fisher's exact in R

```
fisher.test(emp)

# output
Fisher's Exact Test for Count Data

data:  profession and emp.status
p-value = 0.3819
alternative hypothesis: two.sided
...
```

Modelling Count data

- So far we have tested for independence between two (and we can do more) categorical variables using either χ^2 or *Fisher's Exact*
- these tests of independence can be also performed with categorical variables with multple levels (not just binary)
- Later in this module we will model a dependent variable that is binary - **Logistic Regression**

Back to the antibiotic example...

Recall the data:

	short-AB	long-AB
No infection	27	18
Infection	1	4

What would you do to test the hypothesis that there is a connection or relationship between the antibiotics course length and the presence or absence of infection after 6 weeks?

AB in R

```
## Antibiotic Example

```{r}
#inputing the data
ab<-matrix(c(27,18,1,4), nrow = 2)
# Hypothesis Test
fisher.test(ab)

```
```

```
        Fisher's Exact Test for Count Data

data:  ab
p-value = 0.1555
alternative hypothesis: true odds ratio is not equal to 1
95 percent confidence interval:
   0.5186922 305.6258147
sample estimates:
odds ratio
  5.798292
```

What can we conclude?

Why this matters

- Remember how this lecture started: real researchers had all the right numbers - the counts, the percentages - and still got rejected, because they never actually tested their hypothesis
- They had the numbers. They just didn't test them.
- This isn't only a lesson about count data - it's the same gap that shows up whenever descriptive output looks like it has answered the question, but hasn't
- The same trap applies if you take an AI's descriptive summary of your data at face value, without running (and understanding) the actual test behind it

An alternative approach: critical values (optional)

So far we've compared the p-value to α. There's an equivalent way to reach the same conclusion: comparing the test statistic itself to a critical value.

- In order to find the critical value for χ^2 we need:
  - the degrees of freedom
  - the α (significance level) we are testing against
- For a contingency table with r rows and c columns: d.f. $= (r-1) \times (c-1)$
- In our eye/hair colour example: 2 rows, 2 columns, so d.f. $= 1$

Computing the test statistic by hand (optional)

	O	E	$(O-E)^2$	$(O-E)^2/E$
FH & BlueE	38	22.35	244.92	10.96
FH & BrownE	11	26.65	244.92	9.19
DH & BlueE	14	29.65	244.95	8.26
DH & BrownE	51	35.35	244.95	6.93

The test statistic $\chi^2 = 35.34$ is the sum of the last column, and needs to be compared to the critical value from the relevant distribution.

Critical values - back to the eye/hair example (optional)

- If we use $\alpha = 0.05$ (95% confidence level) we can use `qchisq(0.95,1)` to obtain the critical value of 3.84
- In our example the (uncorrected) χ^2 test statistic was 35.34, much larger than the critical value
- We therefore reject H_0 of independence - hair and eye colour are linked
- This is exactly the same conclusion as before, just reached a different way

χ^2 for Goodness of Fit (optional)

- So far we've tested whether two categorical variables are independent. A related test asks a different question: does **one** categorical variable match an **expected** distribution?
- H_0 in this case specifies the **expected (E)** vs **observed (O)** counts in each category
- If we have k categories, the χ^2 statistic is:

$$\chi^2 = \Sigma_{k=1}^{K} \frac{(O_k - E_k)^2}{E_k}$$

- where O_k is the observed frequency for category k, $E_k = p_k \times n$ is the expected frequency under H_0, and this test statistic follows the χ^2_{k-1} distribution with $k-1$ degrees of freedom
- For example: are die rolls from a particular die evenly spread across all 6 faces, as we'd expect if the die is fair?

Effect size for categorical association (optional)

- Just like we used Cohen's d for the difference between two means, categorical association has its own effect size measures
- **Odds ratio**: for a 2x2 table, how much more likely is the outcome in one group compared to the other? An odds ratio of 1 means no association; further from 1 (in either direction) means a stronger association
- `fisher.test()` already reports this - look back at the antibiotics output: an odds ratio around 5.8 means the odds of infection were about 5.8 times higher in one group
- **Cramer's V**: a general-purpose effect size for association between two categorical variables (works for tables larger than 2x2 too), ranging from 0 (no association) to 1 (perfect association)
- As with Cohen's d, a significant p-value tells you *whether* there's an association; the effect size tells you *how strong* it is

What if the samples aren't independent? McNemar's test (optional)

- χ^2 and Fisher's exact both assume the two samples being compared are independent
- Sometimes they're not - for example, testing the same group of patients before and after a treatment (matched pairs)
- **McNemar's test** is designed for this paired/matched case
- In R: `mcnemar.test()`
- The general lesson: as with the t-test (independent vs paired samples), always check whether your categorical data is independent or matched before choosing a test

Non-parametric all along

- You've already used a **non-parametric** test this week, we just hadn't named it that
- χ^2 and Fisher's exact make no assumption about the underlying distribution of a continuous variable - they work directly on counts
- This is the same family of thinking behind other non-parametric methods you'll meet later in the module (and behind bootstrapping, back in Week 2)
- Non-parametric methods are especially useful when: data is categorical to begin with, sample sizes are small, or distributional assumptions (like Normality) don't hold

Testing many pairs of variables? Watch multiple testing (optional)

- If you have many categorical variables and test every pair for independence, some tests will come out "significant" purely by chance
- At $\alpha = 0.05$, testing 20 independent pairs means, on average, one false positive even if nothing is really related
- This is the same issue behind why ablation studies (remember Week 1?) need to account for the number of comparisons being made
- A simple correction: the Bonferroni method - divide α by the number of tests you're running
- Not something to worry about for a single test like today's examples, but essential once you're scanning many variables for relationships

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A statistical gotcha: Simpson's Paradox (optional)

- An association you find in a pooled (combined) table can **reverse or disappear** once you split the data by a third variable
- The classic example: UC Berkeley's 1973 graduate admissions data appeared to show bias against female applicants overall - but broken down department by department, most departments showed no bias, or even favoured women slightly
- The explanation: women disproportionately applied to more competitive departments with lower acceptance rates for everyone
- The lesson: a contingency table's conclusion can depend entirely on what other variable you have (or haven't) accounted for
- Always ask: is there a third variable that could be driving what I'm seeing in this table?

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Where this shows up in the real world (optional)

The same 2x2 (or larger) table logic from today appears constantly in current practice:

- **A/B testing**: did version A or version B of a website lead to more sign-ups? (version $\times$ converted/not)
- **Epidemiology**: vaccine efficacy and exposure studies use exactly this structure (vaccinated/not $\times$ infected/not)
- **Genomics**: case-control studies test whether a genetic variant is associated with a disease
- **Marketing**: is customer churn independent of subscription tier or feature usage?

A very current use: auditing AI systems for fairness

- A live, active use of these exact tests: checking whether a machine learning model's error rates are independent of a protected characteristic (e.g. gender, ethnicity)
- The table looks just like today's examples: *group* $\times$ *model outcome (correct/incorrect, approved/rejected)*
- If the test rejects independence, that's evidence of **disparate impact** - the model's errors aren't evenly spread across groups
- A 100-year-old test (Fisher developed his exact test in the 1920s) is still central to how we check whether today's AI systems are behaving fairly

An AI cautionary tale for this week

- Imagine asking an LLM to "test whether these two categorical variables are related" on a small table, where one or more expected counts are below 5
- An LLM (like a careless human analyst) might default straight to `chisq.test()` without checking that assumption
- As we saw earlier: R won't stop it - it will return a p-value, with only a quiet warning that "the approximation may be incorrect"
- If nobody reads the warning, an unreliable p-value can look just as convincing as a reliable one
- The fix is the same one we've used all lecture: check the expected counts first, and reach for `fisher.test()` when they're small

Week 4 summary

This week after the lecture, the lab and the independent practice you should be:

- work with count data
- able to use R to run table analysis and interpret the results
- next week **Linear Regression**

This week's material is also covered in Chapter 6 p100 - 107 of Crawley Statistics an Introduction using R.